

פתרון 2020 ב מועד 88 – שאלה 7

Expressions	Type Variables
let q = proc (y:?) zero?(-(y,10)) in let p = proc(x: ?) if (x 5) then (q 10) else (q 20) in (p q)	t_f
q	t_q
proc (y:?) zero?(-(y,10))	t_0
let p = proc(x: ?) if (x 5) then (q 10) else (q 20) in (p q)	t_1
p	t_p
proc(x: ?) if (x 5) then (q 10) else (q 20)	t_2
(p q)	t_3
zero?(-(y,10))	t_4
-(y,10)	t_5
if (x 5) then (q 10) else (q 20)	t_6
(x 5)	t_7
(q 10)	t_8
(q 20)	t_9

Expressions	Equations
<pre> let q = proc (y:?) zero?(-(y,10)) in let p = proc(x: ?) if (x 5) then (q 10) else (q 20) in (p q) </pre>	$t_f = t_1$ $t_q = t_0$
<pre> proc (y:?) zero?(-(y,10)) </pre>	$t_0 = t_y \rightarrow t_4$
<pre> let p = proc(x: ?) if (x 5) then (q 10) else (q 20) in (p q) </pre>	$t_1 = t_3$ $t_p = t_2$
<pre> proc(x: ?) if (x 5) then (q 10) else (q 20) </pre>	$t_2 = t_x \rightarrow t_6$
<pre> (p q) </pre>	$t_p = t_q \rightarrow t_3$
<pre> zero?(-(y,10)) </pre>	$t_4 = bool$ $t_5 = int$
<pre> -(y,10) </pre>	$t_5 = int$ $t_y = int$
<pre> if (x 5) then (q 10) else (q 20) </pre>	$t_7 = bool$ $t_6 = t_8 = t_9$
<pre> (x 5) </pre>	$t_x = int \rightarrow t_7$
<pre> (q 10) </pre>	$t_q = int \rightarrow t_8$
<pre> (q 20) </pre>	$t_q = int \rightarrow t_9$

Equations	Substitution
$t_f = t_1$ $t_q = t_0$	$t_7 = bool$
$t_0 = t_y \rightarrow t_4$	
$t_1 = t_3$ $t_p = t_2$	
$t_2 = t_x \rightarrow t_6$	
$t_p = t_q \rightarrow t_3$	
$t_4 = bool$ $t_5 = int$	
$t_5 = int$ $t_y = int$	
$t_6 = t_8 = t_9$	
$t_x = int \rightarrow \mathbf{bool}$	
$t_q = int \rightarrow t_8$	
$t_q = int \rightarrow t_9$	

Equations	Substitution
$t_f = t_1$ $t_q = t_0$	$t_7 = bool$ $t_4 = bool$
$t_0 = t_y \rightarrow \mathbf{bool}$	
$t_1 = t_3$ $t_p = t_2$	
$t_2 = t_x \rightarrow t_6$	
$t_p = t_q \rightarrow t_3$	
$t_5 = int$	
$t_y = int$	
$t_6 = t_8 = t_9$	
$t_x = int \rightarrow bool$	
$t_q = int \rightarrow t_8$	
$t_q = int \rightarrow t_9$	

Equations	Substitution
$t_f = t_1$ $t_q = t_0$	$t_7 = bool$ $t_4 = bool$
$t_0 = \mathbf{int} \rightarrow bool$	$t_y = int$
$t_1 = t_3$ $t_p = t_2$	
$t_2 = t_x \rightarrow t_6$	
$t_p = t_q \rightarrow t_3$	
$t_5 = int$	
$t_6 = t_8 = t_9$	
$t_x = int \rightarrow bool$	

$t_q = int \rightarrow t_8$	
$t_q = int \rightarrow t_9$	

Equations	Substitution
$t_f = t_1$ $t_q = \mathbf{int} \rightarrow \mathbf{bool}$	$t_7 = \mathbf{bool}$ $t_4 = \mathbf{bool}$
$t_1 = t_3$ $t_p = t_2$	$t_y = \mathbf{int}$
$t_2 = t_x \rightarrow t_6$	$t_0 = \mathbf{int} \rightarrow \mathbf{bool}$
$t_p = t_q \rightarrow t_3$	
$t_5 = \mathbf{int}$	
$t_6 = t_8 = t_9$	
$t_x = \mathbf{int} \rightarrow \mathbf{bool}$	
$t_q = \mathbf{int} \rightarrow t_8$	
$t_q = \mathbf{int} \rightarrow t_9$	

Equations	Substitution
$t_f = t_3$	$t_7 = \mathbf{bool}$ $t_4 = \mathbf{bool}$
$t_p = t_2$	$t_y = \mathbf{int}$
$t_2 = (\mathbf{int} \rightarrow \mathbf{bool}) \rightarrow \mathbf{bool}$	$t_0 = \mathbf{int} \rightarrow \mathbf{bool}$
$t_p = (\mathbf{int} \rightarrow \mathbf{bool}) \rightarrow \mathbf{bool}$	$t_q = \mathbf{int} \rightarrow \mathbf{bool}$
$t_5 = \mathbf{int}$	$t_1 = t_3$
$t_6 = t_8 = t_9 = \mathbf{bool}$	$t_x = \mathbf{int} \rightarrow \mathbf{bool}$
$t_q = \mathbf{int} \rightarrow t_8$ $\Rightarrow \mathbf{t_8 = bool}$	
$t_q = \mathbf{int} \rightarrow t_9$ $\Rightarrow \mathbf{t_9 = bool}$	

מהשלב האחרון מתקבל ש-

$$t_f = t_3 = \mathbf{bool}$$